

The Impact of Joint Patent R&D by Domestic and Foreign Investors on the Innovation Capability of Local Enterprises

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Abstract: This paper treats the first joint patent R&D activity between a listed firm and a foreign R&D center as the treatment event, constructs a multi-period difference-in-differences model, and uses data on listed firms from 2010 to 2019 to examine the impact of joint patent R&D on firms' innovation capacity. The results show that, first, joint patent R&D between listed firms and foreign R&D centers significantly reduces the innovation capacity of listed firms, with the decline in independent R&D being particularly pronounced. Second, in terms of mechanisms, after jointly applying for patents with foreign R&D centers, listed firms experience significant declines in both the number of R&D personnel employed and the share of R&D expenditure devoted to independent R&D, providing evidence of a dependence effect. Third, strengthening industry-university-research collaboration, accelerating the development of technology finance, and enhancing intellectual property protection can effectively mitigate the negative impact of joint patent applications on the innovation capacity of listed firms. This study contributes to a better understanding of how high-quality foreign investment affects the innovation of domestic firms and provides a theoretical reference for the design of differentiated innovation-support policies.

Key words: Joint Patent Application; Innovation Capability; Dependence Effect; Foreign R&D Centers

JEL Classification Code: O32; O38; F21

1. Introduction

Technological innovation can give rise to new industries and business models and is an essential driver of rapid economic growth. As an economy develops, factor-driven growth becomes insufficient to meet the requirements of sustainable development, giving way to innovation-driven growth. As the primary carriers of research and development, firms possess distinct advantages in advancing innovation and facilitating the commercialization of its outcomes.

Among the various measures to enhance firms' innovative capacity, learning from advanced foreign technologies is an important channel. For example, domestic firms may engage in joint patent development and application with R&D centers established by foreign-invested enterprises. However, such joint R&D activities may have heterogeneous effects on firms' innovation performance. On the one hand, foreign R&D centers can bring a global perspective as well as advanced technologies and managerial expertise, thereby helping domestic firms improve their innovative capacity through technology demonstration, labor mobility, and upstream-downstream linkages. On the other hand, they may also generate a *dependence effect*, whereby foreign partners dominate the collaboration and domestic firms become increasingly reliant on foreign R&D centers for R&D activities, potentially undermining their capacity for independent innovation.

The existing literature has examined the effects of joint innovation and collaborative R&D on firm innovation and shows that R&D collaboration can indeed improve firm performance (Belderbos et al., 2004; Becker and Dietz, 2024). However, these studies do not specifically focus on how technology support from foreign-invested entities affects local firms' R&D capabilities. Against this backdrop, this paper compiles data on foreign R&D centers established in China, treats joint patenting between Chinese listed firms and foreign R&D centers as the treatment event, and constructs a multi-period difference-in-differences model. Using panel data on listed firms' patenting activities from 2010 to 2019, the paper investigates the impact of joint patenting on the innovation capacity of Chinese listed firms.

This paper makes three marginal contributions. First, it integrates the literature on FDI spillovers with that on joint innovation by examining how joint patenting between foreign-invested and domestic firms affects the innovation capacity of domestic firms, thereby extending the relevant theoretical framework. Second, it focuses on joint patent applications between Chinese firms and

foreign R&D centers. Since foreign R&D centers are primarily engaged in technological innovation, examining them allows for a more precise identification of the technology demonstration effect associated with foreign entry. Third, the paper finds that technological collaboration between domestic firms and foreign R&D centers may have a potentially negative effect on domestic firms' innovation capacity, in the sense that domestic firms may develop a *dependence effect* on foreign partners. This finding provides useful implications for policymakers seeking to optimize the utilization of foreign investment.

The remainder of the paper is organized as follows. Section 2 reviews the literature and discusses the underlying theoretical mechanisms. Section 3 presents the empirical design and data construction. Section 4 reports the main empirical results. Section 5 provides the mechanism analysis. Section 6 offers further analysis. Section 7 concludes and discusses the policy implications.

2. Literature Review and Theoretical Analysis

2.1 Literature Review

2.1.1 The Literature on the Spillover Effects of Foreign Entry.

Early theories of international investment already highlighted the technological spillovers generated by multinational enterprises, arguing that multinationals diffuse advanced technologies into host-country markets and thereby induce local learning effects in technology and management (Hymer, 1976). With the rise of empirical analysis, scholars began to use firm-level data to examine the spillover effects of foreign entry on the innovation, R&D, and other business activities of domestic firms. Some studies construct industry-level measures of foreign entry based on individual foreign investment projects to investigate such spillovers (Javorcik, 2004; Keller and Yeaple, 2009; Xu and Sheng, 2012; Arif-Ur-Rahman and Inaba, 2021). Others exploit FDI liberalization triggered by policy changes, such as revisions to the *Catalogue for the Guidance of Foreign Investment Industries* and the negative list for foreign investment access, as quasi-natural experiments and employ difference-in-differences models (Lu et al., 2017; Hou et al., 2021; Tang et al., 2024). More recent studies have begun to explore the mechanisms underlying FDI spillovers in greater depth, showing that after entering the host country, foreign firms often establish business linkages with only a subset of domestic firms. This feature makes it possible to identify treated firms more

precisely and to construct difference-in-differences models based on foreign entry events (Keller, 2021; Alfaro-Urena et al., 2022; Amiti et al., 2024).

The spillover effects of foreign entry mainly operate through four channels: the demonstration effect, the competition effect, labor mobility, and vertical linkages. The demonstration effect arises when foreign firms introduce new technologies and managerial expertise into the host country, encouraging local firms to engage in imitation and learning. If domestic firms can effectively absorb and internalize such knowledge, their innovative capacity may improve; however, if foreign firms impose substantial barriers to learning or local firms lack sufficient absorptive capacity, the demonstration effect may instead turn negative (Hamida and Gugler, 2009; Jude, 2016; Orlic et al., 2018; Nguyen et al., 2024). With respect to the competition effect, foreign entry exposes local firms to greater competitive pressure, which may generate a crowding-out effect. At the same time, a moderate increase in competition may also force domestic firms to strengthen innovation and enhance their competitiveness (Aitken and Harrison, 1999; Merlevede et al., 2014; Deng et al., 2024). The labor mobility effect operates through knowledge transfer and skill training provided by foreign firms, which improve the host country's human capital. When skilled workers subsequently move to domestic firms, they bring tacit knowledge spillovers that can enhance firms' innovation capacity (Fosfuri et al., 2001; Balsvik, 2011; Liu et al., 2014; Anwar and Sun, 2023). Finally, the vertical linkage effect arises when foreign firms establish upstream and downstream relationships with host-country firms, particularly by becoming high-quality suppliers or customers within supply chains, thereby improving firms' overall performance, including technological innovation and total factor productivity (Javorcik, 2004; Alfaro-Urena et al., 2022).

2.1.2 The Literature on the Determinants of Firm Innovation.

A large body of studies uses indicators such as the number of patent applications, patent grants, and patent citations to measure firms' innovative capacity and to evaluate how various factors affect firm innovation. These factors mainly include the following. First, openness to the outside world, through the attraction of foreign direct investment and the promotion of international technology exchange and cooperation, generates technological spillovers that foster firm innovation (Deng et al., 2024; Gao et al., 2024). Second, talent mobility, especially the movement of highly skilled workers across firms and in particular from foreign-invested firms to domestic firms, facilitates the diffusion of tacit knowledge and the accumulation of managerial experience. This human capital

spillover effect significantly enhances the innovation capacity of domestic firms (Balsvik, 2011; Poole, 2013). Third, policy support, including government R&D funding, tax incentives, and innovation subsidies, promotes firms' innovation investment and strengthens their R&D capabilities (Agrawal et al., 2020). Fourth, industry organization, such as industry associations and industrial alliances, can promote collaborative innovation among firms through information sharing, resource integration, and standard setting (Qiao et al., 2014; Brunner et al., 2024; Bonani, 2025). Fifth, digital technologies, including big data, artificial intelligence, and the Internet of Things, not only provide firms with new R&D tools and innovation resources, but also significantly improve production efficiency and technological innovation capacity (Ancillai et al., 2023).

2.2 Theoretical Analysis

When domestic firms engage in joint patent R&D with foreign R&D centers, different modes of collaboration may exert heterogeneous effects on the innovation capacity of domestic firms, with both positive and negative implications.

2.2.1 Joint patent R&D have positive effect on the innovation capacity of domestic firms.

When domestic firms engage in joint innovation with foreign R&D centers, such collaboration directly embodies a technology demonstration effect. First, foreign R&D centers are predominantly established by investors from developed economies and possess technologies at the global frontier. Through collaboration, domestic firms can gradually learn, absorb, and internalize these technologies, thereby strengthening their own innovative capacity. Second, joint innovation with foreign R&D centers enables domestic firms to acquire more efficient R&D management practices, which can improve internal resource allocation and further enhance innovation performance. Finally, joint innovation between domestic and foreign firms creates a high-level platform for international collaboration, through which domestic firms can broaden their global perspective, better understand worldwide innovation trends, and align their R&D activities more closely with market demand.

2.2.1 Joint patent R&D have negative effect on the innovation capacity of domestic firms.

If collaboration between domestic firms and foreign R&D centers takes the form of contractual outsourcing or intellectual property sharing agreements, other mechanisms may emerge that are detrimental to the improvement of domestic firms' innovation capacity, most notably a *dependence effect* in the innovation process. The dependence effect implies that when specialized institutions

such as foreign R&D centers provide professional innovation and R&D services in the market, local firms may become increasingly reliant on these foreign R&D centers to conduct innovation activities, rather than developing their own in-house R&D capabilities.

The emergence of a *dependence effect* is mainly attributable to two factors. On the one hand, compared with firms in developed economies, Chinese firms in some sectors started later in technological R&D, face higher costs, and still lag behind technologically, which may force them to follow the R&D trajectory set by foreign firms in their innovation activities. Such a constrained innovation path hinders domestic firms from exploring and developing distinctive technologies or products, thereby suppressing their innovative potential. On the other hand, innovation and R&D are business decisions characterized by substantial investment, long horizons, slow returns, and high risk. In an increasingly competitive market environment, entrepreneurs with more conservative management philosophies and stronger risk aversion are more likely to outsource innovation activities to specialized foreign R&D centers that offer lower costs, higher efficiency, and more mature technologies.

However, excessive reliance may cause innovation resources to become concentrated in foreign R&D centers, thereby generating a crowding-out effect. This crowding-out effect can be divided into **product-market crowding out** and **factor-market crowding out**. Product-market crowding out arises because foreign R&D centers possess superior technological capabilities and managerial expertise, giving them a strong competitive advantage in providing innovation-related services such as R&D consulting, technical problem-solving, and technology transfer. Moreover, as foreign R&D centers undertake more commissioned innovation projects, economies of scale further reinforce their technological and cost advantages. As a result, the R&D divisions of domestic firms may find it increasingly difficult to compete in the market for innovation services and may be forced either to exit or to survive only in low-value-added R&D segments. Factor-market crowding out occurs because the high-quality technological services provided by foreign R&D centers are further translated into competitiveness in factor markets. Foreign R&D centers can offer more attractive compensation packages, working conditions, and career prospects, enabling them to attract relatively scarce high-quality R&D and managerial talent. In addition, they often enjoy stronger financial backing and can secure advanced R&D equipment and laboratory facilities through leasing or procurement, ultimately weakening the innovation capacity of domestic firms.

Based on the above analysis, this paper argues that joint patent R&D between domestic firms and foreign R&D centers has a two-sided effect on the innovation capacity of domestic firms. On the positive side, it generates a technology demonstration effect that helps enhance domestic firms' innovation capacity. On the negative side, it may give rise to a *dependence effect*, which is detrimental to the improvement of domestic firms' innovation capacity. To identify the net effect, this paper treats joint patent applications between Chinese domestic listed firms and foreign R&D centers as the treatment event, constructs a multi-period difference-in-differences model, and conducts an empirical analysis.

3 Empirical Design

3.1 Joint Patenting Behavior

This paper focuses on joint patent R&D activities between domestic listed firms and foreign R&D centers. The empirical design is set out as follows.

3.1.1 Collecting Patent Data on Foreign R&D Centers

This paper uses the advanced search function provided by *Tianyancha* to identify foreign-invested firms whose names or company profiles contain keywords such as *R&D Center*, *Research and Development Center*, and related terms¹. During the screening process, foreign firms with no social insurance records are excluded to avoid the inclusion of shell companies. Based on this procedure, the paper identifies 2,432 foreign R&D centers established in China and collects information on their operating status, date of establishment, approval date, industry classification, business address, company profile, and business scope.

The data on foreign R&D centers are further processed in two ways. First, the industry classification is adjusted. The reported industry of foreign R&D centers is typically *Scientific Research and Technical Services*, whereas this paper is more concerned with the types of manufacturing industries to which these centers provide technological services. To address this issue, the paper conducts a text analysis of the company profiles and business scopes of foreign R&D

¹ *Tianyancha* primarily provides professional services for firm-level information retrieval and corporate relationship mapping. Its data are sourced from official institutions, including the National Enterprise Credit Information Publicity System of China, China Judgments Online, and the China National Intellectual Property Administration.

centers and reclassifies each center into the corresponding four-digit manufacturing industry based on the sector it primarily serves. Second, patent information is retrieved. Using the data interface provided by *Tianyancha*, the paper collects patent information for each foreign R&D center, including patent application numbers, application dates, and patent applicants.

3.1.2 Identifying Joint Patenting Behavior

This paper first identifies all domestic listed manufacturing firms from the listed-company database. Using the data interface provided by *Tianyancha*, it then retrieves the complete patent records of each listed firm. The patent databases of these two types of entities are merged and sorted by the unique application number assigned to each patent by the patent office. At this stage, a given patent application number may appear more than once, which can reflect three possible situations: (1) joint patenting among two or more Chinese listed firms; (2) joint patenting among two or more foreign R&D centers; and (3) joint patenting between one or more Chinese listed firms and one or more foreign R&D centers. This paper focuses on the third case.

In the sample, 51 Chinese listed firms are found to have jointly developed patents with foreign R&D centers. The two two-digit industries with the highest incidence of joint patenting are *computer, communication, and other electronic equipment manufacturing*, and *electrical machinery and equipment manufacturing*. The two provincial-level regions with the largest number of such collaborations are Guangdong and Beijing. In addition, 29 listed firms engaged in cross-provincial joint patenting with foreign R&D centers. In terms of patent type, joint patenting is concentrated mainly in utility model patents and invention patents. With respect to the time distribution, joint patenting between listed firms and foreign R&D centers was relatively limited before 2011, began to surge after 2012, and peaked in 2017. Accordingly, this paper sets the study period to 2010-2019, which helps, to some extent, avoid interference from the global financial crisis, the China-US trade conflict, and the COVID-19.

3.1.3 Constructing the Difference-in-Differences Treatment Event

This paper defines domestic listed manufacturing firms that engage in joint patenting as the treatment group, and takes the year in which a treated firm first jointly develops a patent with a foreign R&D center as the treatment year. Domestic listed manufacturing firms that do not engage in joint patenting are assigned to the control group. It should be noted that, although the initial contact between listed firms and foreign R&D centers may precede the completion of a jointly

developed patent, joint patenting serves to identify domestic listed firms with stable collaborative relationships and filters out the uncertainty associated with failed early-stage cooperation. Therefore, using this point in time as the treatment year is well justified.

Moreover, technological innovation and R&D are characterized by long time horizons. When a domestic listed firm and a foreign R&D center jointly conduct R&D and apply for a patent, their interests become closely intertwined. In addition, the process from patent application to patent grant typically takes several years, making short-term collaboration unlikely. Once the patent is granted, both the domestic and foreign parties jointly hold the patent rights, which also implies that their cooperative relationship is likely to persist. Therefore, joint patenting between domestic and foreign entities has lasting effects and can serve as a valid treatment event for examining foreign technology spillovers.

3.2 Multi-period Difference-in-Differences Model

Using domestic listed manufacturing firms as the research sample, this paper treats joint patenting between listed firms and foreign R&D centers as the treatment event and constructs a multi-period difference-in-differences model, as specified in Equation (1).

$$Patent_{scjt} = \alpha + \beta treat_{scjt} + \gamma X_{scjt} + \lambda_s + \delta_{ct} + \mu_{jt} + \epsilon_{scjt} \quad (1)$$

In the model, the subscripts s , t , c , and j denote listed firm, year, city, and industry, respectively.

The dependent variable, $Patent_{scjt}$, captures the innovative capacity of listed firms and is measured by three indicators: (1) the total number of patent applications filed by listed firm s in year t , tpt_{scjt} ; (2) the number of patents independently filed by firm s in year t , tpi_{scjt} ; and (3) the number of patents jointly filed by firm s with other firms in year t , tpj_{scjt} . Patent data for listed firms are obtained from the Chinese Research Data Services Platform (CNRDS), which aggregates each listed firm's patent data into the corresponding categories. Since the dependent variable is a patent count with a large number of zero observations, the paper primarily employs the Poisson pseudo-maximum likelihood (PPML) estimator.

The key explanatory variable in the model, $treat_{scjt}$, is a dummy variable for joint patent R&D, indicating whether listed firm s engages in joint patenting with a foreign R&D center in year t . If $treat_{scjt} = 1$, listed firm s has already established a linkage with a foreign R&D center by year t and has undertaken some form of innovation collaboration. If $treat_{scjt} = 0$, listed firm s has not

yet established such a linkage by year t and has not engaged in innovation collaboration.

In the model, X_{scjt} denotes a set of control variables. Specifically, this paper includes listed firm s ' total profits in year t , $profit_{scjt}$; total assets in year t , $capital_{scjt}$; cash payments for the acquisition and construction of fixed assets, intangible assets, and other long-term assets in year t , $asset_{scjt}$; and employee compensation payable in year t , $wage_{scjt}$, as control variables. These control variables are also obtained from the CNRDS database.

In the model, λ_s , δ_{ct} , and μ_{jt} denote firm fixed effects, city-year fixed effects, and industry-year fixed effects, respectively. ϵ_{scjt} is the error term, with standard errors clustered at the city level.

3.3 Descriptive Statistics

This paper matches listed firms' patenting activities, the first joint patenting event between listed firms and foreign R&D centers, and the firm-level control variables by stock code to construct a panel dataset of listed firms for the period 2010-2019. The resulting sample consists of 40 treated listed firms and 2,392 control listed firms. Descriptive statistics for the main variables are reported in Table 1. All control variables are transformed into logarithms.

Table 1. Descriptive Statistics of the Main Variables

Variable	Observations	Mean	S. D.	Minimum	Maximum
tpt	17 142	44.212	283.622	0	14343
tpi	17 142	34.874	199.442	0	10822
tpj	17 142	9.339	157.87	0	7823
$treat$	17 693	0.013	0.113	0	1
$profit$	17 685	18.801	1.476	10.504	24.797
$capital$	17 684	21.918	1.209	17.019	27.468
$asset$	17 640	18.540	1.69	7.346	24.722
$wage$	17 558	16.895	1.632	4.783	23.063

4 Empirical Results

4.1 Baseline Regression Results

The baseline regressions are estimated using the PPML model to examine the effect of joint patent R&D with foreign R&D centers on the innovation capacity of domestic listed manufacturing firms over the period 2010-2019. The baseline results are reported in Table 2.

Columns (1) and (2) of Table 2 use the number of independently developed patents filed by

listed firms as the dependent variable. Both columns control for time-varying firm characteristics and firm fixed effects. Column (1) additionally includes year, city, and industry fixed effects, and the estimated coefficient on the core explanatory variable is -0.541, significant at the 1% level. Column (2) imposes a more stringent specification by further controlling for city-year and industry-year interaction fixed effects; the estimated coefficient on the core explanatory variable is -0.593, also significant at the 1% level. These results indicate that, after a domestic listed manufacturing firm engages in joint patent R&D with a foreign R&D center, the number of patents it develops independently declines by 41.8%-44.7%.

Columns (3) and (4) of Table 2 instead use the number of jointly developed patents filed by listed firms as the dependent variable, with the same set of controls and fixed effects as in the first two columns. The estimated coefficient on the core explanatory variable is 0.482 in column (3), while the coefficient in column (4) is 0.660 and significant at the 10% level. These results suggest that, after a domestic listed manufacturing firm engages in joint patent R&D with a foreign R&D center, the number of jointly developed patents filed by the listed firm increases by 93.4%.

The above findings yield the following implication. Joint patent R&D between Chinese listed firms and foreign R&D centers may mainly take the form of listed firms outsourcing R&D tasks to foreign R&D centers under intellectual property sharing agreements. In this case, the relationship between listed firms and foreign R&D centers is characterized by a *dependence effect*, which ultimately weakens the firms' capacity for independent R&D. At the same time, under arrangements involving task outsourcing and shared intellectual property rights, the resulting patents are directly counted as jointly developed patents of the listed firms, thereby exerting a positive effect on firms' collaborative innovation activities.

Columns (5) and (6) of Table 2 use the total number of patents filed by listed firms as the dependent variable, with the same set of controls and fixed effects as in the preceding columns. In column (5), the estimated coefficient on the core explanatory variable is 0.084 and statistically insignificant. In column (6), the estimated coefficient is -0.265 and likewise insignificant. These results suggest that joint patent R&D between listed firms and foreign R&D centers has no significant effect on firms' overall innovation capacity, possibly because the negative effect on independently developed patents is offset by the positive effect on jointly developed patents.

Table 2. Effects of Joint Patent R&D on the Innovation Capacity of Domestic Listed Firms

Variables	(1) Independent R&D	(2) Independent R&D	(3) Joint R&D	(4) Joint R&D	(5) Total Patents	(6) Total Patents
<i>treat</i>	-0.541*** (0.133)	-0.593*** (0.157)	0.482 (0.311)	0.660* (0.358)	0.084 (0.277)	-0.265 (0.257)
Controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Year/City/Industry FE	Y		Y		Y	
City-Year FE		Y		Y		Y
Industry-Year FE		Y		Y		Y
Observations	15 162	14 342	10 085	8 734	15 575	14 794
R2	0.8881	0.9163	0.9054	0.9483	0.9051	0.9309

Notes: The regressions are estimated using the PPML model. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Robust standard errors clustered at the city level are reported in parentheses. Unless otherwise stated, the same applies to the tables below. The smaller number of observations in columns (3) and (4) is due to missing data on jointly developed patents in the CNRDS database.

4.2 Robustness Checks

4.2.1 Parallel Trends Test

A key identifying assumption of the multi-period difference-in-differences model is the parallel trends assumption, which requires that, prior to the treatment event, listed firms with and without joint patenting exhibit similar trends in innovation capacity. In other words, the difference between the two groups should remain stable before treatment, or equivalently, their direction and pace of change should be comparable. To test the validity of the multi-period DID design and to examine the dynamic effects of the treatment event, this paper employs an event-study approach to conduct the parallel trends test, as specified in Equation (2).

$$Patent_{scjt} = \alpha + \sum_{k=-10}^{10} \beta D_{scjt}^k + \gamma X_{scjt} + \lambda_s + \delta_{ct} + \mu_{jt} + \epsilon_{scjt} \quad (2)$$

In Equation (2), $D_{scjt}^k = 1[t = \tau_s + k]$ is the interaction term between the multi-period DID treatment indicator *treat* and the event-time dummy for period *k*. Here, τ_s denotes the year in which listed firm *s* first engaged in joint patenting, and $k \in [-10, 10]$. Accordingly, $\tau_s + k$ represents each event year from 10 years before to 10 years after the first joint patent R&D. The period immediately prior to the treatment event ($k = -1$) is omitted as the reference period in the parallel trends test. Using the total number of firm patents as the dependent variable, Figure 1 shows that the estimated coefficients for all pre-treatment periods are statistically insignificant, thereby

supporting the parallel trends assumption. However, after joint patenting begins in the treatment group, no significant effect is observed during the first three years, and a negative effect only emerges starting in the fourth year.

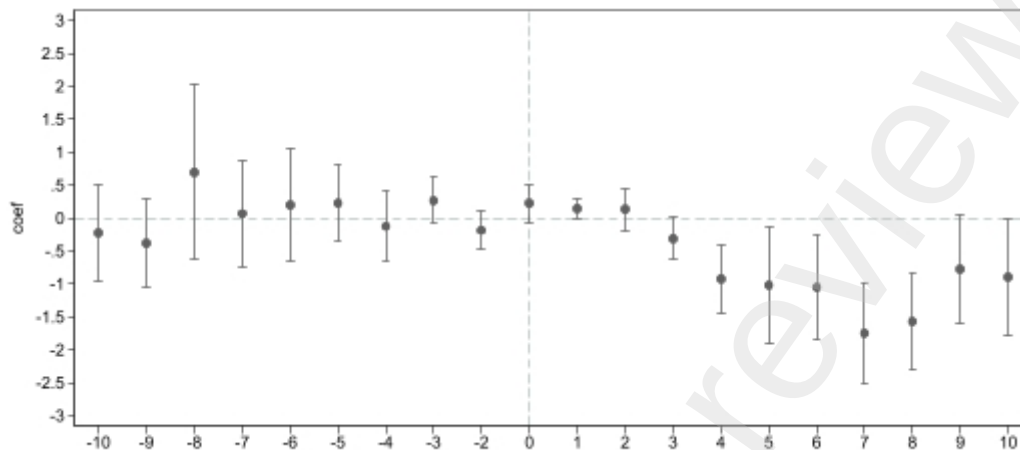


Figure 1. Parallel Trends Test for Total Patents

This paper also conducts parallel trends tests using the number of independently developed patents and the number of jointly developed patents filed by listed firms as dependent variables. The results are presented in Figure 2. It can be seen that, prior to joint patent R&D between the treatment group and foreign R&D centers, the estimated coefficients fluctuate around zero and are statistically insignificant, indicating that the parallel trends assumption is satisfied. As for the long-run effects, once joint patent R&D begins, it immediately and significantly reduces the number of independently developed patents filed by treated firms, and this negative effect persists over time. In addition, in the treatment year, the number of jointly developed patents filed by treated firms initially increases. However, after the third year, the effect turns from positive to negative, and the negative impact remains persistent thereafter. Taken together, the parallel trends result in Figures 1 and 2 support the validity of the multi-period DID design used in this paper. They also suggest that, after listed firms engage in joint patent R&D with foreign R&D centers, their R&D activities become path-dependent, which is, on balance, detrimental to the improvement of their innovation capacity.

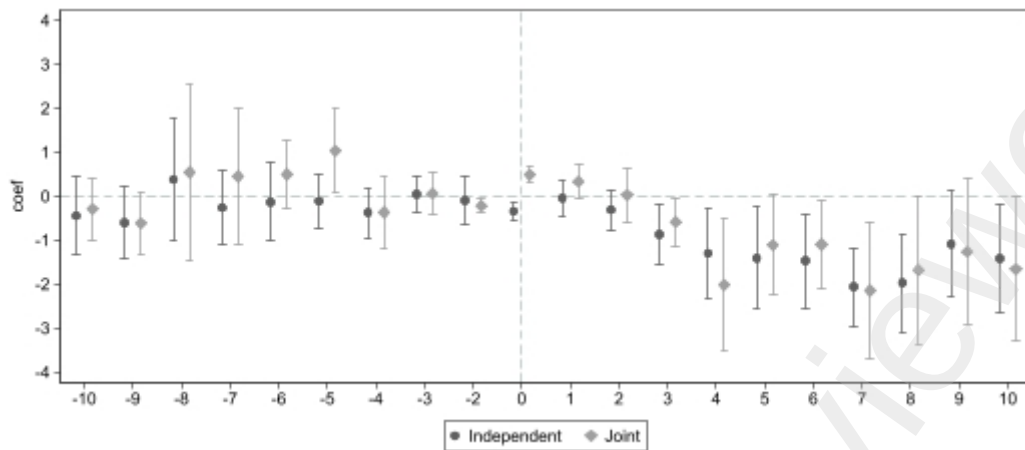


Figure 2. Parallel Trends Tests for Independent R&D and Joint R&D

4.2.2 Random Assignment of the Treatment Group and Treatment Year

To rule out the possibility that the estimated treatment effects in the baseline regressions are merely driven by random patterns in the data, incidental time trends, or other unaccounted-for factors, this paper conducts a placebo test by randomly assigning the treatment group and the treatment year. The random sampling procedure is repeated 1,000 times. The placebo results are reported in Figure 3. In the left panel, the dependent variable is the number of independently filed patents; in the right panel, it is the number of jointly filed patents. For independently filed patents, the baseline estimate of the core explanatory variable is -0.593, which is far below the range of coefficients obtained from the 1,000 random repetitions (and is therefore not marked in the figure). For jointly filed patents, the baseline estimate of the core explanatory variable is 0.660, which exceeds the coefficient values generated in most of the 1,000 repetitions. Both sets of results pass the placebo test, indicating that the estimated effects are indeed driven by the treatment event examined in this paper.

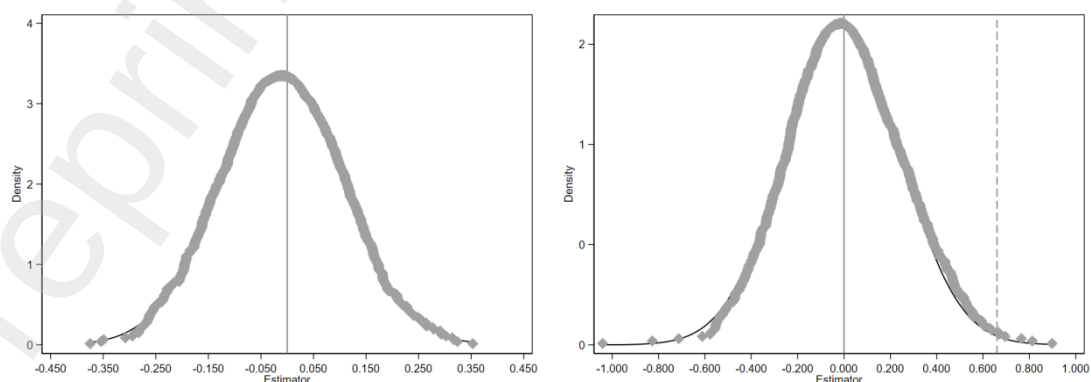


Figure 3. Random Assignment of the Treatment Group and Treatment Year

4.2.3 Heterogeneous Treatment Effects in the Multi-period DID Framework

Traditional DID models are typically based on panel data and examine the effects of a single policy shock by exploiting variation across individuals and over time. In this paper, however, joint patent R&D between listed firms and foreign R&D centers does not occur simultaneously across firms, but is instead adopted gradually over time. Under a multi-period DID framework, although two-way fixed effects (TWFE) can effectively control for firm and time fixed effects, de Chaisemartin and d'Haultfoeuille (2020), Callaway and Sant'Anna (2021), Goodman-Bacon (2021), and Sun and Abraham (2021) show that TWFE may generate substantial bias in estimating the average treatment effect. In the context of this paper, for example, the average treatment effect obtained from the baseline regressions is a weighted average of different treatment effects, consisting of four components: (1) early-treated firms versus later-treated firms; (2) later-treated firms versus early-treated firms; (3) treated firms versus control firms; and (4) treated firms versus always-treated firms. Among these, the treatment effects in the second comparison and the fourth comparison may be biased, thereby affecting the accuracy of the overall treatment effect estimate.

To address this issue, this paper follows Goodman-Bacon (2021) and decomposes the overall treatment effect into the four comparison groups described above. Table 3 reports the decomposition results for the specifications using the number of independently filed patents and the number of jointly filed patents as dependent variables, respectively. The results show that the second and fourth comparison groups, the ones potentially subject to bias, account for only a very small share of the overall treatment effect, with weights of just 0.2% and 0.8%, respectively. This suggests that treatment-effect heterogeneity in the multi-period DID setting is not a major concern in this paper.

Table 3. Goodman-Bacon Decomposition

Decomposition of DID Estimates	Weight	Number of	Number of
		Independently Filed	Jointly Filed
		Patents	Patents
		Average DID Estimate	Average DID Estimate
Early-treated group vs. later-treated group	0.002	-20.668	390.801
Later-treated group vs. early-treated group	0.002	81.009	357.129
Treated group vs. never-treated group	0.988	-0.791	272.214
Treated group vs. always-treated group	0.008	39.703	82.288

Note: The terms before and after vs. denote the treatment group and the control group, respectively.

4.2.4 Alternative Dependent Variables

To ensure that the estimated effects of joint patent R&D are not driven by the specific choice of these indicators, but instead reflect a more general and consistent pattern, the dependent variables are replaced with the number of successfully granted independently filed patents, successfully granted jointly filed patents, and the total number of successfully granted patents, respectively. The regressions include control variables, firm fixed effects, city-year fixed effects, and industry-year fixed effects, and are estimated using the PPML model. The results are reported in Table 4. The estimated coefficients on the core explanatory variable remain consistent with the baseline findings: joint patent R&D has a significantly negative effect on the number of independently granted patents, a significantly positive effect on the number of jointly granted patents, and no significant effect on the total number of granted patents.

Table 4. Effects of Joint Innovation on Patent Grants of Domestic Listed Firms

Variables	(1) Independent Patent Grants	(2) Joint Patent Grants	(3) Total Patent Grants
<i>treat</i>	-0.786*** (0.292)	1.085*** (0.411)	-0.114 (0.310)
Controls	Y	Y	Y
Firm FE	Y	Y	Y
City-Year FE	Y	Y	Y
Industry-Year FE	Y	Y	Y
Observations	14 346	8 008	14 719
R2	0.9095	0.9431	0.9235

4.3 Heterogeneous Industry-level Effects of Joint Patenting

Given differences across industries, this paper classifies sample firms into high-technology industries and low-technology industries based on the 2017 national standard for industrial classification. The detailed classification is reported in Table 5.

To provide a more intuitive view of how joint patent R&D affects the overall innovation capacity of listed firms, this paper uses the total number of patents as the dependent variable and directly applies an event-study approach to illustrate the heterogeneity in long-run effects. The results are presented in Figure 4. It can be seen that, starting from the fourth period after joint patenting begins, the innovation capacity of listed firms in high-technology industries is negatively affected, whereas that of firms in low-technology industries is positively affected. This finding suggests that, once listed firms engage in joint patent R&D with foreign R&D centers, firms in high-technology industries are more likely to develop dependence, thereby weakening their own capacity

for independent R&D.

Table 5. Manufacturing Industry Classification

Category	Industry
Low-technology industries	Agro-food processing (C13); Food manufacturing (C14); Manufacture of liquor, beverages, and refined tea (C15); Tobacco products (C16); Textile industry (C17); Textile, apparel, and accessories manufacturing (C18); Leather, fur, feather and related products, and footwear manufacturing (C19); Timber processing and manufacture of wood, bamboo, rattan, palm, and straw products (C20); Furniture manufacturing (C21); Paper and paper products (C22); Printing and reproduction of recording media (C23); Manufacture of cultural, educational, industrial art, sports, and entertainment goods (C24); Processing of petroleum, coal, and other fuels (C25); Manufacture of chemical raw materials and chemical products (C26); Pharmaceutical manufacturing (C27); Chemical fiber manufacturing (C28); Rubber and plastics products (C29); Non-metallic mineral products (C30); Smelting and pressing of ferrous metals (C31); Smelting and pressing of non-ferrous metals (C32); Metal products (C33); Other manufacturing (C41); Comprehensive utilization of waste resources (C42); Repair of metal products, machinery, and equipment (C43)
High-technology industries	General equipment manufacturing (C34); Special-purpose equipment manufacturing (C35); Automobile manufacturing (C36); Railway, ship, aerospace, and other transportation equipment manufacturing (C37); Electrical machinery and equipment manufacturing (C38); Computer, communication, and other electronic equipment manufacturing (C39); Instrument and meter manufacturing (C40)

This result may be explained by at least two factors. First, there are differences in absorptive capacity. Low-technology industries have lower technological thresholds, so in the process of building innovation linkages, the advanced technologies provided by foreign R&D centers are more easily absorbed and applied by domestic firms, thereby enhancing their innovation capacity. By contrast, high-technology industries involve more sophisticated technological development, which may increase firms' reliance on foreign partners and, in the long run, weaken their capacity for independent innovation.

Second, the explanation may lie in firms' positions within the value chain and the structure of market competition. Low-technology industries are typically located in the lower or middle segments of the industrial chain and tend to be vertically complementary to the major industries of developed economies, making foreign firms more inclined to transfer technology to domestic firms. High-technology industries, however, are more likely to face direct competition from foreign firms. In such cases, foreign firms place greater emphasis on protecting intellectual property and core technologies, often granting Chinese listed firms only limited rights to use the technology, thereby making it more difficult for them to learn technologies in key domains.

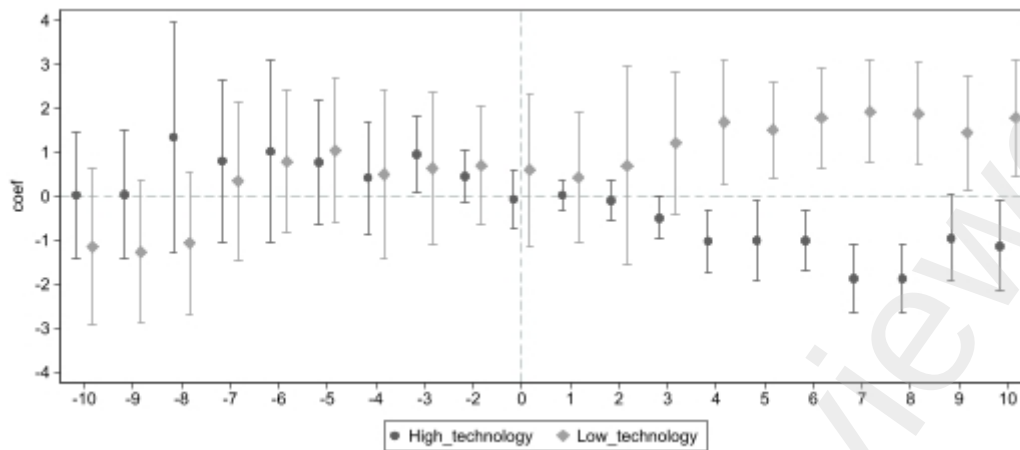


Figure 4. Heterogeneous Effects of Joint Patent R&D Across Industry Types

4.4 Ownership Relationships

During the process of joint patent application, some listed firms acquire equity stakes in foreign R&D centers. To exclude the potential confounding effect of ownership relationships between listed firms and foreign R&D centers, while further examining the heterogeneous effects associated with such relationships, this paper checks whether an ownership link exists between each listed firm and the corresponding foreign R&D center involved in joint patenting and constructs a dummy variable, $holding_{st}$. If listed firm s establishes an ownership relationship with the corresponding foreign R&D center in year t , $holding_{st}$ is assigned a value of 1, otherwise, it is assigned a value of 0.

The dummy variable $holding_{st}$ and its interaction term with the multi-period DID treatment variable are jointly added to the baseline model, and the results are reported in **Table 5**. The estimated coefficient on the core explanatory variable remains negative for the number of independently filed patents and positive for the number of jointly filed patents, consistent with the baseline results. The interaction-term estimates further show that, for listed firms that establish ownership links with the corresponding foreign R&D centers, independent R&D capacity declines further, while joint R&D activity increases further, ultimately leading to an increase in the total number of patents. A possible explanation is that equity participation strengthens the patent-related R&D collaboration between the two parties, thereby amplifying its impact on the innovation capacity of listed firms.

Table 6. Listed Firms' Equity Participation in Foreign R&D Centers

Variables	(1) Independent R&D	(2) Joint R&D	(3) Total Patents
$treat \times holding$	-0.515** (0.219)	0.179 (0.557)	0.545** (0.219)
<i>treat</i>	-0.471** (0.191)	0.485 (0.480)	-0.491*** (0.180)
<i>holding</i>	0.060 (0.117)	0.321 (0.305)	0.006 (0.209)
Controls	Y	Y	Y
Firm FE	Y	Y	Y
City-Year FE	Y	Y	Y
Industry-Year FE	Y	Y	Y
Observations	14 342	8 734	14 794
R2	0.916	0.948	0.931

5 Mechanism Analysis

To further explore the mechanisms through which joint patent R&D affects the innovation performance of listed firms, and to better understand why it may have a negative effect on firms' independent innovation, this paper examines the impact of joint patenting on firms' R&D personnel and R&D investment.

5.1 Number and Share of R&D Personnel

When listed firms engage in joint patent R&D with foreign R&D centers, the number of R&D personnel employed by the listed firms may decline. Foreign R&D centers are specialized R&D institutions capable of undertaking a large volume of outsourced R&D projects and improving efficiency through economies of scale. By establishing innovation partnerships with foreign R&D centers, listed firms may rely on their technological capabilities and avoid duplicative or basic R&D tasks. As part of this collaborative process, some R&D activities are shifted to foreign R&D centers, reducing the need for listed firms to maintain their original number of R&D personnel and leading them to cut redundant R&D positions in order to lower labor costs.

This paper tests whether joint patent R&D between listed firms and foreign R&D centers leads to a reduction in the number of R&D personnel employed by listed firms, thereby contributing to a decline in their overall innovation capacity. Data on the number of R&D personnel and the share of R&D personnel in domestic listed firms are obtained from the CNRDS database and used as

dependent variables. Since the number and share of R&D personnel are not count variables, the regressions are estimated using OLS, controlling for firm-level characteristics, firm fixed effects, and both city-year and industry-year interaction fixed effects. The results are reported in Table 7. The estimated coefficients on the core explanatory variable are negative in both specifications. While the effect on the number of R&D personnel is not statistically significant, the effect on the share of R&D personnel is significant at the 5% level, indicating that joint patent R&D reduces the share of R&D personnel in listed firms by 1.9%.

Table 7. Number and Share of R&D Personnel

Variables	(1) Number of R&D Personnel	(2) Share of R&D Personnel
<i>treat</i>	-5.109 (3.288)	-0.019** (0.009)
Controls	Y	Y
Firm FE	Y	Y
City-Year FE	Y	Y
Industry-Year FE	Y	Y
Observations	9 237	9 232
R2	0.970	0.918

This paper also uses the number of R&D personnel in domestic listed firms as the dependent variable and applies an event-study approach to examine the long-run effect of joint patent R&D on firms' R&D staffing. Since the R&D personnel data available from the CNRDS database cover only the period 2013-2019, the event study is conducted over a six-period window before and after the treatment event. The results are presented in Figure 5. The estimated coefficients exhibit a clear shift from positive to negative after the onset of joint patent R&D, with the coefficient in the first post-treatment year being significantly negative at the 5% level. This result suggests, to some extent, that joint patenting increases listed firms' reliance on foreign technology in R&D, thereby reducing their internal stock of R&D personnel.

A reduction in R&D personnel may destabilize the core technology teams of listed firms and lead to the loss of R&D talent. If listed firms become overly dependent on external technologies and resources in joint patent R&D, their internal R&D teams may be marginalized or gradually dissolved, leaving R&D personnel with fewer opportunities to solve problems independently and engage in technological breakthroughs. Over time, firms may lose their capacity to compete in future

independent R&D, gradually surrender control over core technologies and innovation trajectories, and find it difficult to build sustained technological competitiveness.

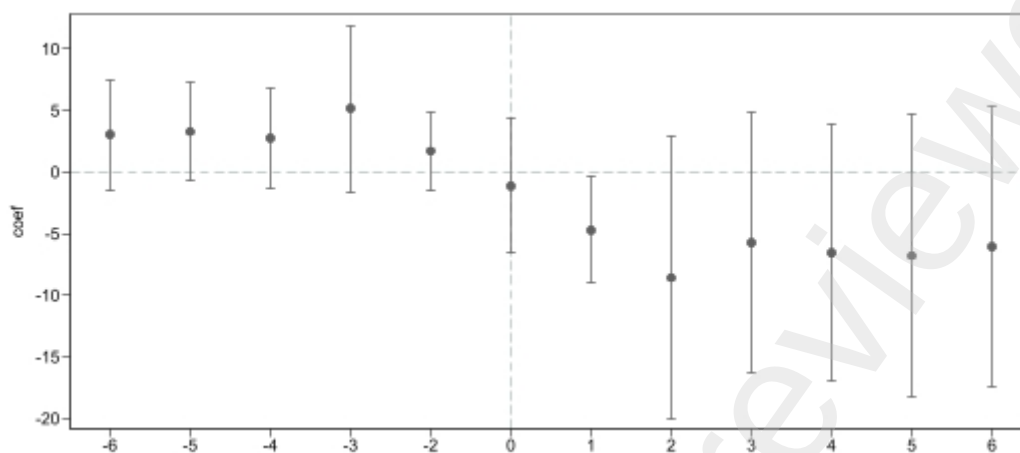


Figure 5. Long-run Effect of Joint Patent Applications on Firms' Number of R&D Personnel

5.2 Structural Adjustment in R&D Investment

Once joint patent R&D takes place, listed firms may outsource part of their R&D activities to foreign R&D centers and substitute external technologies and resources for in-house research, thereby affecting their own R&D investment. This paper first uses the amount of R&D expenditure and the ratio of R&D expenditure to operating revenue for domestic listed firms as dependent variables in a multi-period DID analysis. The data are also drawn from the CNRDS database. The results are reported in Table 8. The estimated coefficients of joint patent R&D on both the amount of R&D expenditure and its share in operating revenue are statistically insignificant.

Table 8. R&D Expenditure and Share

Variables	(1) R&D Expenditure	(2) R&D Expenditure as a Share of Operating Revenue
<i>treat</i>	1.692 (1.407)	-0.003 (0.005)
Controls	Y	Y
Firm FE	Y	Y
City-Year FE	Y	Y
Industry-Year FE	Y	Y
Observations	15 103	15 103
R2	0.947	0.634

To examine the long-run effect of joint patent R&D on R&D investment, this paper uses the amount of R&D expenditure by domestic listed firms as the dependent variable and employs an

event-study approach. The results are presented in Figure 6. The estimated coefficients are statistically insignificant throughout the six periods before and after the treatment event, and no clear trend is observed.

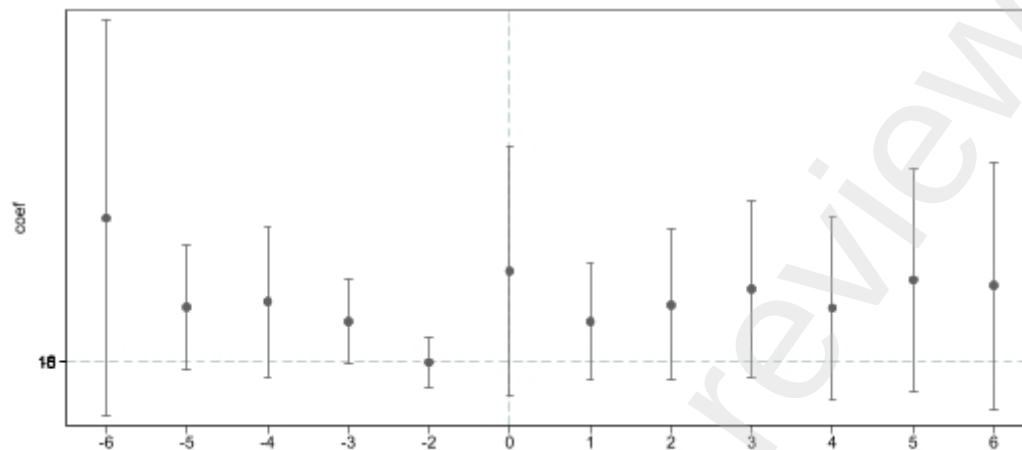


Figure 6. Long-run Effect of Joint Patent Applications on Firms' R&D Investment

Although neither the amount of R&D expenditure nor its share in firm profits changes significantly, joint patent R&D may affect the internal composition of R&D investment. Firm R&D investment can be broadly divided into two categories: expensed R&D and capitalized R&D. Expensed R&D refers to R&D outlays that are directly recognized in the income statement in the current period, including short-term, consumable, and non-capitalizable expenditures, such as outsourcing service fees, routine administrative expenses, material costs, short-term project expenses, and labor costs. Capitalized R&D, by contrast, refers to expenditures recorded on the balance sheet and amortized over future periods, involving long-term R&D investment expected to generate future economic benefits, such as investment in fixed assets and intangible assets. In the case of joint patent R&D between listed firms and foreign R&D centers, the dominant pattern is that listed firms outsource R&D tasks to foreign R&D centers, which generates substantial coordination and service-related expenses during the collaboration period; these expenditures are therefore mostly classified as expensed R&D. By contrast, the need for in-house R&D by listed firms declines accordingly, reducing demand for long-term capital investment such as R&D equipment and infrastructure, and thereby lowering capitalized R&D.

Based on this distinction, this paper uses expensed R&D and capitalized R&D of domestic listed firms as dependent variables and examines them with an event-study approach. The results are shown in Figure 7. After joint patent R&D begins, expensed R&D investment by listed firms is

positively affected. Although the coefficients in most years do not reach significance at the 5% level, they exhibit a clear upward trend, suggesting increasing economic significance. By contrast, the estimated coefficients for capitalized R&D are mostly negative and display a declining trend. These results indicate, to some extent, that joint patent R&D increases listed firms' expensed R&D outlays while reducing their capitalized R&D spending. Thus, although total R&D investment does not change significantly, its internal composition appears to undergo structural adjustment.

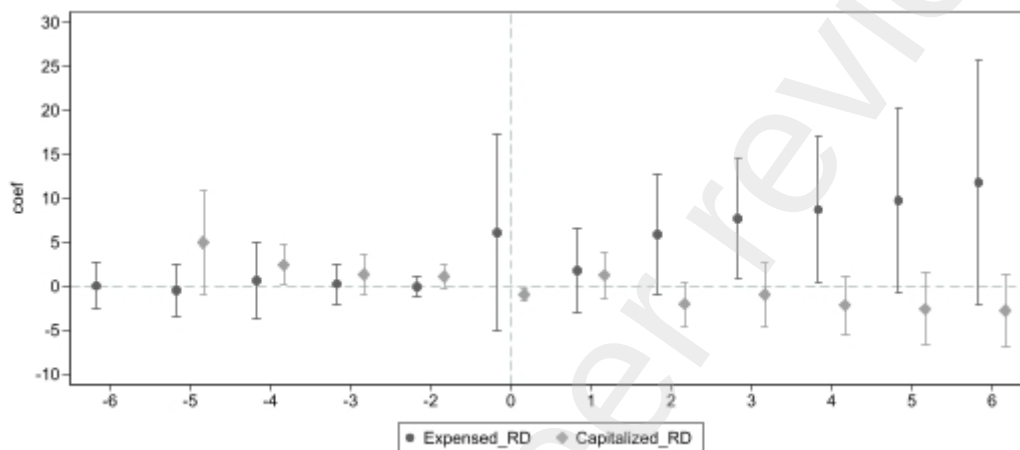


Figure 7. Effects of Joint Patent Applications on Expensed and Capitalized R&D Investment

To provide more direct evidence on how joint patent R&D affects the internal reallocation of listed firms' R&D expenditure, this paper uses detailed firm-level R&D expense data from the CNRDS database for the period 2017-2019. It searches the expense-description field using keywords such as *commissioned*, *outsourced*, *cooperation*, *coordination*, *collaboration*, *joint*, *intermediary*, and *technical services*. R&D expenses containing these keywords are classified as collaborative R&D expenses, while all remaining R&D expenses are classified as non-collaborative R&D expenses. The shares of these two categories in total R&D expenditure are then calculated and used separately as dependent variables in the multi-period DID model. The results are reported in Table 9. The findings show that joint patent R&D indeed increases the share of collaborative R&D expenses while reducing the share of non-collaborative R&D expenses. Since non-collaborative R&D expenses mainly include expenditures such as R&D workers' salaries and R&D equipment investment, this shift may weaken the future innovation capacity of listed firms, thereby further confirming the baseline findings of this paper.

Table 9. Collaborative and Non-collaborative Innovation Expenditures

Variables	(1)	(2)
	Share of Collaborative Innovation Expenditure	Share of Non-collaborative Innovation Expenditure
<i>treat</i>	0.033* (0.018)	-0.035* (0.019)
Controls	Y	Y
Firm FE	Y	Y
City-Year FE	Y	Y
Industry-Year FE	Y	Y
Observations	6 104	6 104
R2	0.910	0.901

6 Mitigating the Dependence Effect of Joint Patent R&D

Joint patent R&D between listed firms and foreign R&D centers may have a negative effect on the development of listed firms' own R&D capabilities. In the further analysis, this paper focuses on moderating factors that may influence the above negative effect, with the aim of making better use of foreign technology to enhance domestic innovation capacity.

6.1 Promoting Industry-University-Research Collaborative Innovation

The concept of industry-university-research collaborative innovation in China can be traced back to the 1980s and gradually became an important development consensus with the advance of market-oriented reforms and the rise of emerging industries. Through such collaboration, firms can access technologies, talent, and equipment from universities and research institutes-resources that are often difficult for firms to obtain on their own. This can shorten the R&D cycle, reduce investment costs, and improve innovation efficiency. Against this backdrop, this subsection examines whether improving the environment for industry-university-research collaboration can enhance the innovation capacity of listed firms.

Because listed firms are generally large enterprises within their regions, they are more likely to be affected by improvements in the local environment for industry-university-research collaboration. This paper uses two province-level indicators from the Report on China's Regional Innovation Capacity for 2010-2019, namely knowledge acquisition (f1) and innovation environment (f2), to capture the regional environment for industry-university-research collaborative innovation. The knowledge acquisition indicator includes sub-indicators such as

scientific and technological cooperation, technology transfer, and investment by foreign-funded firms. The innovation environment indicator covers sub-indicators such as innovation infrastructure, the market environment, labor quality, the financial environment, and entrepreneurship. Taken together, these sub-indicators reflect, from different perspectives, the quality of the local environment for industry-university-research collaborative innovation.

$$Patent_{scjt} = \alpha + \beta_1 treat_{scjt} \times f_{pt} + \beta_2 treat_{scjt} + \gamma X_{scjt} + \lambda_s + \delta_{ct} + \mu_{jt} + \epsilon_{scjt} \quad (3)$$

This paper interacts the treatment variable capturing joint patent R&D between listed firms and foreign R&D centers with the regional indicators of the industry-university-research collaborative innovation environment, and incorporates the interaction terms into the baseline regression model to estimate Equation (3). In the model, the subscripts s , t , c , p , and j denote listed firm, year, city, province, and industry, respectively. The results are reported in Table 10. It can be seen that, regardless of whether the dependent variable is the total number of patents, the number of independently developed patents, or the number of jointly developed patents filed by listed firms, the estimated coefficients on the interaction terms are all significantly positive. These findings suggest that the better the local environment for industry-university-research collaborative innovation, the more it helps enhance firms' innovation capacity, thereby mitigating the negative effects associated with joint patent R&D with foreign R&D centers.

Table 10. Tests Based on Sub-indicators from the Evaluation of China's Regional Innovation

	Capacity					
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	Total Patents	Independent R&D	Joint R&D	Total Patents	Independent R&D	Joint R&D
$treat \times f1$	0.681*** (0.116)	0.553*** (0.104)	0.482* (0.269)			
$treat \times f2$				0.352*** (0.105)	0.171** (0.072)	0.867*** (0.288)
$treat$	-0.863*** (0.188)	-1.069*** (0.140)	0.253 (0.379)	-0.586*** (0.225)	-0.749*** (0.137)	-0.075 (0.430)
Controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
City-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y	Y	Y
Observations	14 794	14 342	8 734	14 794	14 342	8 734
R2	0.931	0.916	0.948	0.931	0.916	0.948

Furthermore, this paper more directly examines how listed firms' innovation capacity is affected when they engage in joint R&D with universities and research institutes. To this end, the paper collects all patent application data for listed firms and identifies patents jointly applied for with universities, academies, and research institutes based on applicant information; these are defined as industry-university-research collaborative patents. The first year in which a listed firm engages in such collaboration is treated as a new event shock, and a multi-period DID variable, *treat_cxy_{scjt}*, is constructed accordingly. Both joint patent R&D with foreign R&D centers and industry-university-research collaboration are then incorporated into the same econometric model. The estimation results are reported in Table 11.

It can be seen that, regardless of whether the dependent variable is the total number of patents, independently developed patents, or jointly developed patents, the estimated coefficient on the first occurrence of industry-university-research collaboration is consistently positive. The coefficients are significantly positive at the 1% level for total patents and jointly developed patents. These results indicate that collaboration with domestic universities and research institutes can enhance the innovation capacity of listed firms through channels such as technology demonstration. This further suggests that strengthening industry-university-research collaboration helps mitigate the weakening effect on innovation associated with joint patent R&D with foreign R&D centers.

Table 11. Test of Industry-University-Research Collaboration by Listed Firms

Variables	(1) Total Patents	(2) Independent R&D	(3) Joint R&D
<i>treat</i>	-0.399* (0.212)	-0.663*** (0.163)	0.242 (0.269)
<i>treat_cxy</i>	0.266** (0.107)	0.182 (0.131)	0.672*** (0.207)
Controls	Y	Y	Y
Firm FE	Y	Y	Y
City-Year FE	Y	Y	Y
Industry-Year FE	Y	Y	Y
Observations	14 794	14 342	8 734
R2	0.931	0.916	0.949

6.2 Supporting the Development of Technology Finance

Innovation and R&D require substantial financial investment and are characterized by long time horizons and considerable uncertainty. Technology finance integrates technological innovation

with financial services by supporting technological development and innovative activities through financial instruments. On the one hand, technology finance helps firms obtain the funding they need through channels such as venture capital, bank loans, and government funding. On the other hand, it can direct capital toward high-potential technology projects, optimize resource allocation, accelerate the commercialization of innovation outcomes, and improve overall innovation efficiency. This subsection adopts two approaches to examine the impact of technology finance development on the innovation capacity of listed firms.

First, this paper uses the establishment of technology-finance pilot cities as a proxy for the level of technology finance development in the city where a listed firm is located. Technology-finance pilot cities were jointly designated by the Ministry of Science and Technology, the People's Bank of China, and other government agencies. The first batch included 16 cities nationwide, such as Beijing and Shanghai, and the coverage was further expanded in subsequent years. This paper constructs a dummy variable, SD_c , indicating whether city c is a technology-finance pilot city. If city c is designated as such, $SD_c = 1$; otherwise, $SD_c = 0$. Following Equation (3), this dummy variable is interacted with the multi-period DID variable and added to the model. The results are reported in columns (1)-(3) of Table 12.

Table 12 shows that, on the one hand, the estimated coefficient on joint patent R&D between listed firms and foreign R&D centers is significantly negative in all specifications. On the other hand, the coefficient on the interaction term is positive, but statistically significant only for total patents and jointly developed patents. This suggests that, after a city introduces more comprehensive financial policies to support innovation, local listed firms receive stronger financial backing, which helps support their own R&D activities and enhance their innovation capacity. At the same time, the results indicate that the construction of technology-finance pilot cities has a stronger effect on collaborative innovation. A possible explanation is that technology-finance pilot policies are formulated as broad city-level initiatives and thus lack clear targeting and specificity, causing funds to flow more readily into commissioned R&D activities with higher short-term efficiency.

Second, this paper examines government-guided funds as a specific policy instrument of technology finance in order to further explore the moderating effect of technology finance development. Listed firms may actively use government-guided funds to support R&D activities in key technological fields and thereby reduce R&D risk. The paper obtains data on government-

guided funds for 2010-2019 from the Zero2IPO database, including information on establishment dates, whether the managing institution is state-owned, and the registered location. After excluding funds with missing key variables, the paper calculates the number of newly established government-guided funds in each city and year, denoted by $fund_{ct}$, and uses this variable as a proxy for city-level technology finance development. The interaction term between $fund_{ct}$ and the multi-period DID variable is then included in the estimation model. The results are reported in columns (4)-(6) of Table 12.

The estimated coefficients on the interaction term remain positive in all specifications. However, in this case, statistical significance appears only for independently developed patents, at the 5% level. This result suggests, to some extent, that government-guided funds are primarily intended to support breakthroughs in firms' core technologies. As a result, in terms of innovation support, such funds are more likely to flow into independent innovation activities that are conducive to core technological breakthroughs. Consequently, in the context of joint patent R&D between listed firms and foreign R&D centers, government-guided funds exert a positive moderating effect on firms' capacity for independent R&D.

Table 12. Test of Technology Finance Development in the Cities Where Listed Firms Are Located

Variables	(1) Total Patents	(2) Independent R&D	(3) Joint R&D	(4) Total Patents	(5) Independent R&D	(6) Joint R&D
$treat \times SD$	0.821* (0.439)	0.478 (0.407)	1.652*** (0.453)			
$treat \times fund$				0.013 (0.010)	0.022** (0.009)	0.009 (0.020)
$treat$	-1.058*** (0.379)	-1.055*** (0.386)	-0.857*** (0.321)	-0.290 (0.252)	-0.638*** (0.164)	0.648* (0.360)
Controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
City-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y	Y	Y
Observations	14 794	14 342	8 734	14 794	14 342	8 734
R2	0.931	0.917	0.949	0.931	0.916	0.948

6.3 Strengthening Intellectual Property Protection

A sound intellectual property protection regime has an important impact on firms' innovative

behavior. In cities with stronger intellectual property protection, listed firms are more willing to invest in R&D and engage in innovation activities. Once a firm's innovative outcomes are patented, they receive effective legal protection, which reduces the risk of infringement. However, when listed firms engage in joint patent R&D with foreign R&D centers, a stricter intellectual property protection regime may constrain technology spillovers from foreign R&D centers, thereby hindering the improvement of listed firms' independent R&D capacity.

To examine the moderating role of the intellectual property protection regime in the effect of joint patenting between domestic and foreign firms, this paper treats a city's designation as a National Intellectual Property Model City as the treatment event and constructs a dummy variable, IPP_{ct} . If city c is designated as a National Intellectual Property Model City in year t , then $IPP_{ct} = 1$; otherwise, $IPP_{ct} = 0$. This variable is used as a proxy for the development of the local intellectual property protection regime. It is then interacted with the multi-period DID variable for joint patent R&D and incorporated into the regression model. The results are reported in columns (1)-(3) of Table 13. The findings show that, under a stricter intellectual property protection regime, joint patent R&D between listed firms and foreign R&D centers further reduces the total number of patents filed by listed firms, mainly because their independent R&D activity declines even more. This result suggests that, under stronger intellectual property protection, listed firms that rely on external R&D capabilities receive fewer technology spillovers from joint patent R&D.

In the analysis of the intellectual property protection regime, the expected conclusion that stronger protection would encourage listed firms to engage in R&D and innovation is not supported. A possible explanation is that listed firms face financial constraints, which may weaken their incentives for independent R&D even when patent rights are well protected, leading them instead to seek cooperation with specialized R&D institutions. To examine this possibility, this paper follows Hadlock and Pierce (2010) and obtains the SA index of listed firms from the CSMAR database. Since all firms in the sample are listed companies and thus relatively large in size, a larger SA index is interpreted as indicating weaker financing constraints. The SA index is then further interacted with the term $treat_{scjt} \times IPP_{ct}$ to construct a triple interaction term, which is included in the model together with the corresponding two-way interaction terms. The results are reported in columns (4)-(6) of Table 13.

The estimated coefficients on the triple interaction term are all positive. In particular, for

independent R&D by listed firms, the coefficient is significant at the 1% level, substantially mitigating the negative effect associated with joint patent applications with foreign R&D centers. This finding suggests that listed firms facing weaker financing constraints have more abundant financial resources and therefore stronger incentives to undertake independent R&D activities. In addition, the estimated coefficient on the SA index itself is significantly negative for joint R&D by listed firms, implying that firms with weaker financing constraints are more inclined to conduct independent R&D rather than collaborative R&D.

Table 13. Test of the Intellectual Property Protection Regime

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	Total Patents	Independent R&D	Joint R&D	Total Patents	Independent R&D	Joint R&D
$treat \times IPP$	-0.381** (0.180)	-0.326*** (0.080)	0.171 (0.125)	-0.472 (0.296)	-0.587*** (0.135)	0.028 (0.222)
$treat \times IPP \times SA$				0.939 (0.644)	1.482*** (0.386)	1.797* (1.012)
$treat$	0.015 (0.291)	-0.356** (0.150)	0.593 (0.365)	-0.649 (0.627)	-0.842* (0.482)	-0.007 (0.431)
SA				0.399 (0.711)	0.301 (0.692)	-3.664** (1.691)
Other Interactions	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
City-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y	Y	Y
Observations	14 794	14 342	8 734	14 794	14 342	8 734
R2	0.931	0.917	0.948	0.932	0.917	0.949

7 Conclusion and Implications

To examine the impact of joint patent R&D with foreign R&D centers on the innovation capacity of domestic listed firms in China, this paper uses panel data on Chinese listed firms from 2010 to 2019 and employs a multi-period difference-in-differences model, treating joint patent R&D between listed firms and foreign R&D centers as the treatment event to evaluate how this mode of collaboration affects the innovation capacity of domestic firms.

The results show that joint patent R&D has complex effects on the innovation capacity of domestic firms. First, in the short run, joint patent applications significantly reduce the number of

independently developed patents while increasing the number of jointly developed patents; in the long run, however, they may lead to a persistent decline in firms' patenting activity. Second, after engaging in joint patent R&D with foreign R&D centers, domestic firms experience a significant decline in both the number of R&D personnel employed and the share of R&D expenditure devoted to independent R&D, reflecting the presence of a *dependence effect*. If this dependence effect is not properly addressed, it may give rise to a *crowding-out effect* and weaken the innovation capacity of domestic firms. Third, measures such as stronger industry-university-research collaboration, the development of technology finance, and improved intellectual property protection can, to some extent, mitigate the negative effects of joint patent R&D on firm innovation.

The findings of this paper yield the following implications. First, a long-term policy mechanism should be established to support the independent innovation of domestic firms. Through measures such as fiscal subsidies and tax incentives, policymakers should seek to balance collaborative R&D and independent R&D so as to prevent the emergence of a *dependence effect* and even a *crowding-out effect*. Second, greater effort should be devoted to fostering a social and institutional environment conducive to innovation by domestic firms, including stronger industry-university-research collaboration, the development of technology finance, and enhanced intellectual property protection. Third, policy design should fully take firm heterogeneity into account, with policy measures tailored to the specific characteristics of firms across different industries.

One limitation of this paper is that the proxy event used to identify joint patent R&D between domestic listed firms and foreign R&D centers is joint patent application. However, joint patent application may capture different underlying arrangements, including genuine joint R&D as well as commissioned R&D. Although this study has made substantial efforts to distinguish among these cases through manual identification, it has not yet been able to separate them with full precision. This issue remains to be addressed in future research.

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